

```
*****S* Seq: 0x43EE1111 Ack: 0x0 Win: **** TcpLen: 20
[Xref => http://doc.url.net/2016936]

[**] [1:2016936:2] Suspicious inbound to Database port 1433
[**]
[Classification: Potentially Bad Traffic] [Priority: 2]
07/07-10:10.838622 <IP>:<PORT> -> <IP>:<PORT>
TCP TTL:113 TOS:0x0 ID:256 Iplen:22 DgmLen:21
*****S* Seq: 0x6D5F1111 Ack: 0x0 Win: **** TcpLen: 20
[Xref => http://doc.url.net/2016936]
```

From the Snort IDS generated alert file, the records of 'Potentially Bad Traffic' can be cut and placed in a separate CSV file.

The finally obtained CSV file is read in TensorFlow for deep learning of the model and further predictive analysis of upcoming network traffic. Using this approach, the upcoming traffic can be analysed for the probability of being malware or not.

```
>>> import pandas
>>> dataset = pandas.read_csv('mydataset.csv')
>>> dataset.shape
(1000, 5)
>>> data.columns
Index(['Classification', 'DGMLEN', 'IPLLEN', 'TTL',
      'IP'], dtype='object')
>>> y, X = train[['Classification'], train[['DGMLEN', 'IPLLEN',
      'IP']]].fillna(0)
>>> X_Modeltrain, X_Modeltest, y_Modeltrain, y_Modeltest =
train.Modeltest_split(X, y, test_size=0.1, random_state=29)
>>> lr = LogisticRegression()
>>> lr.fit(X_Modeltrain, y_Modeltrain)
>>> print accuracy_score(y_Modeltest, lr.predict(X_
Modeltest))
Output - 0.60183838
```

The accuracy score of existing model can be evaluated

Visualisation of learning and graphs using TensorBoard

The visualisation of learning behaviour can be analysed

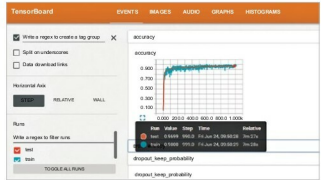


Figure 3: Live demo of TensorBoard

using TensorBoard, which is a set of Web based applications. TensorBoard includes the visualisation of five types of data including images, audio, scalars, graphs and histograms so that better and more effective analysis can be done.

A live demo of TensorBoard is available at <http://www.tensorflow.org/tensorboard/>.

To open and execute TensorBoard, the application should be opened in a Web browser on Port 6006 - <http://localhost:6006/>.

At the top panel, there are navigation tabs using which different types of visualisation options can be selected.

Using TensorFlow for research and development

TensorFlow can be used by research scholars and academicians to implement research proposals and projects that include deep learning. The results of TensorFlow can be compared with other tools for analytics and machine learning so that a clear and self-executed experience can be gained. 

By: Dr Gaurav Kumar

The author is the MD of Magma Research and Consultancy. He is associated with various academic institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools. Email: kumargaurav.in@gmail.com; URL: www.gauravkumarindia.com

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